

Emil Stroecker

Email: es4485@columbia.edu • Cell: +4915168111711 • [Website](#) • [LinkedIn](#) • [GitHub](#) • [ORCID](#)

EDUCATION

Maastricht University

Research MSc in Cognitive Neuroscience — (current GPA: 8.8/10.0 → US: 4.0 /4.0)

Maastricht, Netherlands
09/2024 – 08/2026

- Selective 2-year program with research focus, featuring 10-month MSc thesis research
- *Selected coursework:* Statistics I-III, Applied Statistics I-II, Basic Mathematical Methods, Signal Analysis, MATLAB Advanced Programming, Python Programming, Quantificational Logic and Deductions, Advanced fMRI, EEG & MEG, Brain Connectivity & Connectomics, Sensorimotor Processing, Functional Neuroanatomy

Kyoto University

Kyoto University International Education Program — (US GPA: 4.0/4.0)

Kyoto, Japan
09/2023 – 02/2024

- Received [JASSO scholarship](#)

Maastricht University

BSc in Psychology — Summa Cum Laude, Top 1% (GPA: 9.1/10.0 → US: 4.0/4.0)

Maastricht, Netherlands
09/2021 – 08/2024

- Empirical programme with a focus on research methods, statistics, cognitive psychology, and neuroscience

RESEARCH EXPERIENCE

Columbia University, Zuckerman Institute

MSc Thesis Research Intern at Nikolaus Kriegeskorte's [Visual Inference Lab](#)

New York City, USA
11/2025 – 08/2026

- Project lead: researching neural network models of ventral stream and their bias towards geometrical interpretations of depth in the Ames Window Illusion
- Comparing contemporary neural networks (Metric3D, DSINE, MoGe) for depth estimation
- Running psychophysics experiment (Prolific x PsychoJS) for benchmarking
- Creating online visual perception tasks for large-scale data collection
- Experiments contributed to development of AI psychophysics benchmark tool ("illugen")
- First author for [2-page abstract](#), accepted to Conference on Cognitive Computational Neuroscience 2026

Maastricht University

Independent Researcher under Rainer Goebel on fMRI-BCI letter speller ("RetinoType")

Maastricht, Netherlands
09/2025 – 08/2026

- Project lead: decoding retinotopic representations of imagined letters in striate cortex from 7T fMRI data
- Reconstructing visual field via imagery-pRF forward model to simulate 26-letter imagery
- Developing a simulated proof-of-concept letter speller BCI augmented with a probabilistic model combining confusion matrix with LLM language prior for denoising responses

Research Assistant at [DCN Group](#)

09/2025 – 11/2025

- Refactored and debugged a [psychophysics experiment](#) in PsychoPy for EEG compatibility and improved task paradigm

Sokendai, National Institute for Physiological Sciences

Internship in primate research at Ryo Sasaki's [Multisensory Integration Systems Lab](#)

Okazaki, Japan
07/2025

- Highly selective internship covering comprehensive training on major practical aspects of macaque experiments
- Observed monkey experiments full-time for two weeks, witnessing full procedures for behavioural training & neural recordings for 2AFC saccade tasks & spatial navigation tasks

Maastricht University

BSc Honours Thesis ([MaRBL](#)) under Vincent van de Ven

Maastricht, Netherlands
11/2023 – 07/2024

- Project lead with only minimal supervision: experimental design, experiment creation (PsychoPy/PsychoJS), recruiting, administration to 40 participants, analysis, etc.
- Examined effect of structural association of tones on tone sequence recognition
- Consistent with literature, musicians demonstrated significantly better recognition of structured tone sequences
- Presented final poster at symposium to faculty and fellow honours students, project graded outstanding (10/10)

Interdisciplinary Research Project through [Honours+](#)

11/2022 – 05/2023

- Self-directed, interdisciplinary, empirical research, selecting top 5% of BSc students across disciplines
- Led a team of 5 honours students: conducted a qualitative content analysis on TikTok content, laying foundation for quantitative cross-sectional study with 100+ participants; findings presented to faculty

University Hospital Cologne

Research Intern in TrainParC-Advanced research project

Cologne, Germany
07/2023 – 09/2023

- Conducted and evaluated various neuropsychological & motor tests (MMST, BDI-II, MoCA, FOG-Q, etc.) with Parkinson's patients
- Developed migration from paper to digital data entry pipeline in Excel, significantly reducing workflow time
- Contributions acknowledged in a resulting Springer Nature Link publication ([Trials, 2025](#))

ABSTRACTS & PRESENTATIONS

Conference Abstracts

Stroecker, E., Cheng, F., Kriegeskorte, N. (2026). Measuring Depth Inversion in the Ames Window Illusion [Extended abstract]. 9th Annual Conference for Cognitive Computational Neuroscience, NYC, USA. [\[Link\]](#)

Poster Presentations

- Stroecker, E., van de Ven, V. (2024, July 05). *Measuring on-line predictive recall of structured versus non-structured tone sequences*. [Poster presentation]. *15th Marble Symposium*, Maastricht, The Netherlands. [\[Link\]](#)
- Stroecker, E., Kud, H., Husarcikova, V., van Kuilenburg, K., Eckstein, H. (2023, May 10). *The Impact of TikTok Consumption on the Perception of ADHD*. [Poster presentation]. *Honours+ Closing Symposium*, Maastricht, The Netherlands. [\[Link\]](#)

SCIENCE COMMUNICATION

- Stroecker, E. (2026, June 22). *Beauty and the Brain* [Popular science article]. BrainMatters. [\[Link\]](#)
- Stroecker, E. (2025, October 18). *fNIRS* [Popular science article]. BrainMatters. [\[Link\]](#)
- Stroecker, E. (2025, September 30). *The revolutionary finding that changed the meaning of functional magnetic resonance imaging forever* [Popular science article]. BrainMatters. [\[Link\]](#)
- Stroecker, E. (2025, May 29). *How Brain Reading is Becoming Reality Through Real-Time fMRI* [Popular science article]. BrainMatters. [\[Link\]](#)

HONOURS & AWARDS

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|--|-------------------|
| MSc Honours Programme, PREMIUM — extracurricular business consultant for Atos | 01/2025 – 05/2025 |
| Summa Cum Laude Designation, BSc Psychology (= <u>top 1%</u> of the class) | 08/2024 |
| BSc Honours Programme, MaRBLc (selective for <u>top 20%</u>) — empirical thesis research project | 11/2023 – 07/2024 |
| Scholarship, Japanese Student Services Organization (JASSO) — 400,000JPY = 2,500USD | 10/2023 – 02/2024 |
| BSc Honours Programme, Honours+ (selective for <u>top 5%</u>) — extracurricular research project on societal topics | 11/2022 – 05/2023 |
| BSc Honours Programme, FPN Honours Programme (selective for <u>top 5%</u>) — extracurricular coursework | 09/2022 – 09/2023 |

SERVICE & OTHER

Conference Committees

- Committee Member at [CCN 2027](#) Proceedings Committee Starting: 10/2026
- Committee Member at [CCN2026](#) Technical Program Committee 10/2025 – 08/2026
- Managing and facilitating submission process for 600+ submissions across Proceedings and Extended Abstracts
 - Documenting and deciding conference policies and procedures, shaping future iterations of the venue

Teaching & Mentoring

- Research Mentor at Columbia University for [BUMP Biology](#) 01/2026 – 12/2026
- Advising undergraduate student on first steps into neuroscientific research, i.e. applying for research programmes, contacting PIs, shaping academic goals & development, etc.
- Teaching Assistant at Maastricht University 11/2024 – 07/2025
- Instructed tutorial group meetings and conveyed course content to 60+ undergraduate students
 - Courses: Functional Neuroanatomy, Consciousness, Statistics, Psychodiagnostics

Science Communication

- Editorial Assistant for Editorial Management Team at In-Mind ([in-mind.org](#)) 05/2026 – ongoing
- Spearheading development of editorial guidelines for incoming, peer-reviewed submissions
 - Performing copy-editing on new articles, ensuring accessible and cohesive flow, structure, phrasing, etc.
- Writer for BrainMatters ([brainmatters.nl](#)) 04/2025 – ongoing
- Writing neuroscience-related articles intended for non-expert audiences spanning different topics [\[overview\]](#)

SKILLS

Languages

German (native) | English (professional) | Japanese (intermediate) | Dutch, French, Spanish (elementary)

Applied Skills

- Languages: Python, Bash, Matlab, R, JavaScript, SQL, HTML/CSS
- ML & Data: PyTorch, NumPy, Pandas, Scikit-learn, Matplotlib
- Tools: PsychoPy, BrainVoyager, Blender, LaTeX (Overleaf), GitHub, Slurm, MS Office, OpenReview, Prolific, Qualtrics
- Neuroimaging: fMRI paradigms (real-time fMRI, neurofeedback, ...), EEG, NIBS (TMS, tDCS,...), electrode recording procedures
- Other: Psychophysics, experimental design, academic writing, science communication

Trainings & Certifications

TOEFL iBT 118/120 $\Leftrightarrow$ 6.0/6.0 (equivalent to CEFR C2) | Columbia University ZI fMRI Certification (L2) | UM Honours Professional & Personal Workshops (10+)